

BASICS FOR AIR TRAFFIC CONTROL – PILOT WEATHER REPORTS (PIREPs)

INTRODUCTION

No observation is timelier than the one made from the cockpit. In fact, aircraft in flight are the only means of directly observing cloud tops, icing, and turbulence. Pilots, as well as controllers, welcome Pilot Weather Reports (PIREPs) because they inform pilots of weather where there are no weather reporting stations.

As a controller, you will be handling PIREPs, whether solicited or volunteered, on a daily basis. PIREPs are a valuable source of information to the Air Traffic Service (ATS). You must be able to decode and process them properly and in a timely manner to be of use to the pilot.

The purpose of this module is to introduce you to the PIREP Program operated by the National Weather Service (NWS) and the Federal Aviation Administration (FAA). You will learn to solicit, record, handle, and decode PIREPs.



INTRODUCTION TO PILOT WEATHER REPORTS (PIREPs)

Purpose: The purpose of this lesson is to identify the purpose and uses of pilot weather reports and the information included in the report.

Objectives:

- Identify purpose and uses of a PIREP
- Identify content included in a PIREP

References for this lesson are as follows:

- FAA Order JO 7110.65, Air Traffic Control
- FAA Order JO 7110.10, Flight Services
- Aeronautical Information Manual (AIM)
- FAA-H-8083-28, Aviation Weather Handbook
- FAA Form 7110-2 (PIREP)

Pilot Weather Report (PIREP) In-Flight Reports

A PIREP is a report of meteorological conditions in flight. Since a meteorological phenomenon may not be observable by any other means, pilots can report actual weather conditions as they are encountered by the aircraft in flight.

Example of a PIREP

“ACADEMY APPROACH, SOUTHWEST TWO THIRTY SEVEN 10 MILES SOUTHWEST OF THE APPLETON VORTAC AT 8,000, THE TOPS OF THE BROKEN LAYER WERE 6,000.”



PIREP Classifications

There are two classifications of PIREPs:

- Urgent (UUA)
- Routine (UA)

Urgent (UUA)	Weather phenomena reported by a pilot that represent a hazard or a potential hazard to flight operations. Conditions Classified as Urgent PIREPs ■ Tornadoes, funnel clouds, or waterspouts ■ Severe or extreme turbulence (including clear air turbulence) ■ Severe icing ■ Hail ■ Low-level wind shear ■ Volcanic eruptions and volcanic ash clouds ■ Any other weather phenomena reported that are considered as being hazardous, or potentially hazardous to flight operation
Routine (UA)	All received PIREPs are classified as routine except those classified as urgent.

Reported Conditions

Reported conditions may include, but are not limited to:

- Thunderstorms and related phenomena
- Clouds
 - Bases
 - Tops
 - Layers
- In-flight visibility
- Restrictions to visibility
 - Haze
 - Smoke
 - Dust
- Precipitation
- Wind at altitude
- Temperature aloft
- Airframe icing
- Turbulence
- Wind shear
- Clear air turbulence
- Volcanic ash



PIREP Uses

The following ground stations receive, format, and disseminate information reported in PIREPs.

Air Traffic Control Tower (ATCT)		ATCTs and TRACONs use PIREPs to: ■ Expedite traffic flow in the vicinity of the airport ■ Provide hazardous weather avoidance procedures
Flight Service Stations (FSSs)		FSSs use PIREPs from en route aircraft to: ■ Brief pilots ■ Issue in-flight advisories ■ Issue weather avoidance information
Air Route Traffic Control Center (ARTCC)		ARTCCs use the reports within the center's area to: ■ Expedite the flow of en route traffic ■ Determine most favorable altitudes ■ Issue hazardous weather information
National Weather Service (NWS)		NWS uses PIREPs to: ■ Verify or amend conditions contained in forecasts and advisories triggering mechanism for issuing advisories ■ Pilot weather briefings ■ Study meteorological phenomena
Terminal Radar Approach Control (TRACON), Air Traffic Control *(ATC), Air Traffic Control Tower (ATCT)		All ATC facilities and NWS: ■ Forward PIREPs into the weather distribution system ■ Make PIREP information available to pilots and other interested parties



Knowledge Check A

REVIEW what you have learned so far about PIREPs. ANSWER each question listed below.

1. If reported on PIREP, which of these weather conditions would be classified UUA? (Select all correct answers that apply.)
 - ☒ Severe icing
 - ☒ Volcanic ash
 - ☒ Low-level wind shear
 - ☒ Hail
2. Who uses PIREPs to expedite traffic flow in the vicinity of the airport and to provide hazardous weather avoidance procedures? (Select the correct answer.)
 - ☐ FSS
 - ☒ ATCTs and TRACONS
 - ☐ ATC and NWS
 - ☐ NWS

Soliciting PIREPs



Requirements to Solicit PIREPs

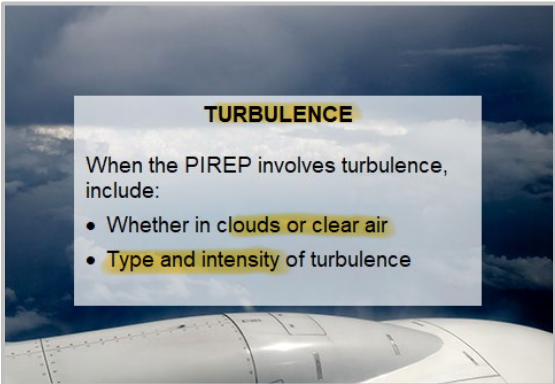
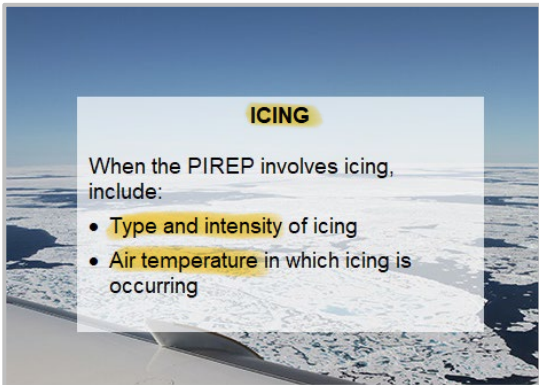
Air traffic facilities are required to solicit PIREPs from aircraft when one of the following conditions **exists** or is **forecasted**:

- | | | |
|---|--|---|
| <ul style="list-style-type: none"> ■ Ceilings at or below 5,000 feet <ul style="list-style-type: none"> • Includes cloud bases/tops when feasible ■ Visibility 5 miles or less <ul style="list-style-type: none"> • Surface or aloft ■ Thunderstorms and related phenomena | <ul style="list-style-type: none"> ■ Turbulence <ul style="list-style-type: none"> • Moderate degree or greater ■ Icing <ul style="list-style-type: none"> • Light degree or greater ■ Wind shear | <ul style="list-style-type: none"> ■ Volcanic ash clouds are reported or forecast ■ Detection of sulfur gases (SO₂), associated with volcanic activity, in the cabin ■ When braking action advisories are in effect (terminal only) |
|---|--|---|



S - Ceilings ≤ 5000
 S - Visibility
 T - Thunderstorms
 W - Wind Shear
 I - Icing ≥ Light
 T - Turbulence ≥ Moderate
 S - Sulphur Gas
 V - Volcanic Ash
 W - When Braking Action

PIREP Preparation

■ Ensure each PIREP includes: • Aircraft position • Time • Altitude or flight level • Type aircraft • At least one other element	PIREP you request: ■ Obtain directly from the pilot Requested by another facility: ■ Pilot transmits directly to that facility
	

Handling PIREPs

Handle PIREPs as follows:

All ATC	All ATC specialists shall relay pertinent PIREP information to concerned aircraft in a timely manner.
En Route ATC	En route ATC specialists shall relay all <u>operationally significant</u> PIREPs to the facility <u>Weather Coordinator</u>. The Weather Coordinator is an en route position that ensures the sectors in the center, towers, and TRACONS are kept up-to-date on significant weather that may affect their areas of responsibility.
Terminal ATC	Terminal ATC specialists relay all operationally significant PIREPs to: ■ Appropriate intra-facility positions ■ FSS serving the area in which the report was obtained ■ Other concerned terminal or en route ATC facilities, including non-FAA facilities

PIREP Form

The PIREP form (FAA Form 7110-2) is used to aid the controller in collecting all data needed to complete a PIREP.

The elements listed on the PIREP form are called **Text Element Indicators (TEIs)**.

Text Element Indicators (TEIs)

PIREP FORM	
Pilot Weather Report	
3-Letter SA Identifier	
1. UA → Routine Report UUA → Urgent Report	
2. /OV →	Location:
3. /TM →	Time:
4. /FL →	Altitude/Flight Level:
5. /TP →	Aircraft Type:
Items 1 through 5 are mandatory for all PIREP's	
6. /SK →	Sky Cover:
7. /WX →	Flight Visibility and Weather:
8. /TA →	Temperature (Celsius):
9. /WV →	Wind:
10. /TB →	Turbulence:
11. /IC →	Icing:
12. /RM →	Remarks:

FAA Form 7110-2 (1-05) Supersedes Previous Edition

FAA Form 7110-2

The FAA, NWS, and other organizations that enter PIREPs into the weather reporting system use the following standardized format.

- Items 1 through 5 are included in all transmitted PIREPs, along with one or more of items 6 through 12

3- or 4-LETTER STATION IDENTIFIER: Nearest weather reporting to the reported phenomenon

REPORT TYPE: Routine or urgent PIREP

LOCATION

TIME: Coordinated Universal Time (UTC)

ALTITUDE: Essential for turbulence and icing reports

TYPE AIRCRAFT: Essential for turbulence and icing reports

PIREP FORM

Pilot Weather Report

1 Letter SA Identifier → UA → Routine Report → UUA → Urgent Report

2 /OV → Location

3 /TM → Time

4 /FL → Altitude/Flight Level

5 /TP → Aircraft Type

Items 1 through 5 are mandatory for all PIREPs

- Items 6 through 12 provide a description of reported weather

SKY COVER: Cloud height and coverage (sky clear, few, scattered, broken, or overcast) and cloud tops

VISIBILITY AND WEATHER: Flight visibility, precipitation, restrictions to visibility, etc.

TEMPERATURE: Degrees Celsius

WIND: Direction in degrees magnetic north and speed in knots

TURBULENCE: LGT, MOD, SEV, EXTRM

ICING: TRACE, LGT, MOD, SEV

REMARKS: For reporting elements not included or to clarify previously reported items

PIREP FORM

6 /SK → Sky Cover

7 /WX → Flight Visibility and Weather

8 /TA → Temperature (Celsius)

9 /WV → Wind

10 /TB → Turbulence

11 /IC → Icing

12 /RM → Remarks

FAA Form 7110-2 (1-05) Supersedes Previous Editions

Note: Information from these forms is entered into the **Aeronautical Information Systems Replacement (AISR)** system or from **Flight Service Stations (FSS)**.



Knowledge Check B

REVIEW what you have learned so far about PIREPs. ANSWER each question listed below.

1. Air traffic facilities must solicit PIREPs when which of the following are reported? *(Select all correct answers that apply.)*
 - ☒ Ceiling at or below 5,000 feet
 - ☐ Trace of ice
 - ☒ Visibility of 5 miles or less
 - ☐ Light turbulence
2. What is the responsibility for all ATC specialists when handling a PIREP? *(Select the correct answer.)*
 - ☒ Relay pertinent PIREP information to concerned aircraft in a timely manner
 - ☐ Relay all operationally significant PIREP information to appropriate intra-facility positions
 - ☐ Relay pertinent PIREP information to FSS serving the area in which the report was obtained
 - ☐ Ensure sectors are kept up-to-date on significant weather that may affect their area of responsibility
3. Which of the following items are required when preparing a FAA Form 7110-2? *(Select all correct answers that apply.)*
 - ☒ Report type
 - ☒ Location
 - ☐ Temperature
 - ☒ Time
 - ☒ 3- or 4-letter station identifier
 - ☒ Type aircraft

Introduction to Pilot Weather Reports Summary

Pilot weather reports serve as a valuable source of information to the ATC community. As a controller, you will handle pilot weather reports on a daily basis. You must be able to decode and process them properly in a timely manner.

PIREP RECORDING AND DECODING

Purpose: The purpose of this lesson is to explain and interpret each segment of information found on a pilot weather report.

Objective:

- Identify information recorded in a PIREP

References for this lesson are as follows:

- FAA Order JO 7110.65, Air Traffic Control
- Aeronautical Information Manual (AIM)

PIREP Recording and Decoding

Classification

Classification of a PIREP refers to the type of message displayed in a report. There are two message types:

- UA – Routine PIREP
- UUA – Urgent PIREP

3-Letter SA Identifier

1. UA → X Routine Report UUA → Urgent Report

Location

Location in reference to a very high frequency (VHF) navigational aid (NAVAID) or an airport using the three or four alphanumeric identifier.

- If appropriate, encode the identifier, then three digits to define a radial and three digits to define the distance in nautical miles
- Route segment with two or more fixes to describe a route

UA /OV APE230010 /TM 1516 /FL085/TP BE20 /SK BKN-TOP065/WX FV03SM BR/TA 02 /VV 23538KT/TB LGT/IC TRACE RIME/RM TCU W-NE	
2. /OV →	Location: 10 miles SW Appleton VORTAC

Examples

/OV SAV-PZD (Savannah, GA, to Albany, GA) or /OV KSAV-KPZD

Time

The time that the phenomenon occurred or was encountered is reported in four digits Coordinated Universal Time (UTC).

UA /OV APE230010 /TM 1516 /FL085/TP BE20 /SK BKN-TOP065/WX FV03SM BR/TA 02 /VV 23538KT/TB LGT/IC TRACE RIME/RM TCU W-NE	
3. /TM →	Time: 1516 zulu

Altitude/Flight Level

Altitude is reported from where the phenomenon was first encountered in hundreds of feet.

- If not known, enter UNKN (unknown)
- If the aircraft was climbing or descending:
 - **DURC** – During climb in remarks
 - **DURD** – During descent in remarks

*C : Climb
DURC
D : Descent
DURD*

UA /OV APE230010/TM 1516 /FL085/TP BE20 /SK BKN-TOP065/WX FV03SM BR/TA 02 /VV 23538KT/TB LGT/IC TRACE RIME/RM TCU W-NE	
/FL →	Altitude/Flight Level: 8,500

Examples

/FLUNKN (altitude unknown); **/RM DURC** (Remarks: during climb)

If the condition was encountered within a layer, the altitude range appears within the appropriate Text Element Indicator (TEI) describing the condition.

Type Aircraft

If type aircraft is not known, the contraction UNKN is reported.

- Icing and turbulence reports must always include the type aircraft

UA /OV APE230010/TM 1516 /FL085/TP BE20 /SK BKN-TOP065/WX FV03SM BR/TA 02 /VV 23538KT/TB LGT/IC TRACE RIME/RM TCU W-NE	
/TP →	Aircraft Type: King air 200

Examples

/TP C172 (Cessna 172); **/TP PAY2** (Piper Cheyenne 2)

Sky Condition

Height of cloud bases, tops, and cloud coverage are reported as follows:

- Cloud cover abbreviation (FEW, SCT, etc.)
- Height of cloud bases, using three digits, in hundreds of feet mean sea level (MSL) (if unknown, use UNKN or for bases, leave blank)
- Height of tops in hundreds of feet, preceded by the word "TOP"
- When more than one layer is reported, use a solidus (/) to separate layers

UA /OV APE230010/TM 1516 /FL085/TP BE20
/SK BKN-TOP065/WX FV03SM BR/TA 02
/WV 23538KT/TB LGT/IC TRACE RIME/RM TCU W-NE

6. /SK → Sky Cover: **Tops of the broken layer were 6,500**

Examples

/SK BKN-OVC UNKN TOP060/BKN120 TOP150/ SKC – indicates the base is a broken to overcast layer, altitude unknown, top of the first layer at 6,000 MSL. Second layer base at 12,000 MSL tops 15,000 MSL, clear above.



Knowledge Check C

REVIEW what you have learned so far about PIREPs. ANSWER each question listed below.

1. What would a proper classification of a PIREP be? *(Select the correct answer.)*
 - ☐ UC
 - ☒ UA
 - ☐ MT
2. Which example of time reported on a PIREP is correct? *(Select the correct answer.)*
 - ☐ /1200
 - ☐ /0800 TM
 - ☒ /TM 1600
3. Which example represents sky condition reported on a PIREP? *(Select the correct answer.)*
 - ☒ /SK BKN-TOP065
 - ☐ /OV APE230010
 - ☐ /WV 23538KT

Weather

Weather conditions encountered by the pilot are reported as follows:

- **Flight visibility**, if reported, will be the first entry in the /WX field
 - Enter **FV** followed by a two-digit visibility value rounded down, if necessary, to the nearest whole statute mile and append "SM" (Example: **FV03SM**)
 - If visibility is reported as **unrestricted**, it will be entered as **FV99SM**

UA /OV APE230010/TM 1516 /FL085/TP BE20	
/SK BKN-TOP065/WX FV03SM BR/TA 02	
/VV 23538KT/TB LGT/IC TRACE RIME/RM TCU W-NE	
7. /WX →	Flight Visibility and Weather: <i>Flight visibility is 3 miles, mist</i>

Air Temperature

Outside air temperature is reported using two digits in degrees Celsius.

- Negative temperatures are prefixed with an "M" (Example: **M32**)

UA /OV APE230010/TM 1516 /FL085/TP BE20	
/SK BKN-TOP065/WX FV03SM BR/TA 02	
/VV 23538KT/TB LGT/IC TRACE RIME/RM TCU W-NE	
8. /TA →	Temperature (Celsius): 02

Examples

The PIREP Form 7110-2 indicates to write negative temperatures with a hyphen. When PIREPS are disseminated, negative temperatures will be indicated with an M (i.e.: -2 = M02).

Wind

Wind is reported using three digits to indicate wind direction (magnetic) and two to three digits to indicate reported wind speed and is **always appended with "KT."**

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UA /OV APE230010/TM 1516 /FL085/TP BE20  
/SK BKN-TOP065/WX FV03SM BR/TA 02  
/WV 23538KT/TB LGT/IC TRACE RIME/RM TCU W-NE
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9. /WV →	Wind: 23538KT
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Examples

/WV 26580KT (wind two six five at eight zero); /WV 235110KT (wind two three five at one one zero);
/WV 05008KT (wind zero five zero at eight)

Turbulence

Turbulence is reported with duration, intensity, type, and altitude as follows:

- Intensity using contractions **LGT**, **MOD**, **SEV**, or **EXTRM**
 - Range or variation of intensity will be separated with a hyphen (Example: **MOD-SEV**)
- CAT (Clear Air Turbulence) or CHOP will be included if reported by the pilot
- Altitude will be reported only if it differs from value reported in /FL
 - When a layer of turbulence is reported, height values will be separated with a hyphen
 - If lower or upper limits are not defined, **BLO** (below) or **ABV** (above) will be used

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UA /OV APE230010/TM 1516 /FL085/TP BE20  
/SK BKN-TOP065/WX FV03SM BR/TA 02  
/WV 23538KT/TB LGT/IC TRACE RIME/RM TCU W-NE
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10. /TB →	Turbulence: LGT
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Examples

/TB MOD (Moderate turbulence); /TB EXTRM CAT (Extreme Clear Air Turbulence); /TB LGT-MOD 040-085 (Light to moderate turbulence between four thousand and eight thousand five hundred); /TB INTMT LGT CHOP (Intermittent light chop)

Icing

Intensity, type, and altitude of icing are reported as follows:

- Intensity is reported first using the word **TRACE** or the contractions **LGT**, **MOD**, or **SEV**
 - Range or variation of intensity reports are separated with a hyphen
 - Icing forecasted, but not encountered, is reported using **NEG**
- Icing type is reported as **RIME**, **CLR**, or **MX**
- Icing/altitude reported only if different from the value reported in the /FL

When icing is reported, always report temperature in the /TA Text Element Indicator (TEI).

UA /OV APE230010/TM 1516 /FL085/TP BE20 /SK BKN-TOP065/WX FV03SM BR/TA 02 /WV 23538KT/TB LGT/ IC TRACE RIME /RM TCU W-NE	
11. /IC →	Icing: Trace RIME

Examples

/IC LGT RIME (Light rime icing); **/IC MOD MX** (Moderate mixed icing)

Remarks

Remarks are used to report phenomena that are considered important but do not fit in any of the other areas.

- Hazardous weather is reported first
- Low-Level Wind Shear (LLWS) is described to the extent possible

Remarks include but are not limited to:

- | | |
|-------------------------|--|
| ■ LLWS reports | ■ Dust storms |
| ■ Thunderstorm lines | ■ Sandstorms |
| ■ Coverage and movement | ■ Clouds observed, but not encountered |
| ■ Hail | ■ Contrails |
| ■ Lightning | |

UA /OV APE230010/TM 1516 /FL085/TP BE20 /SK BKN-TOP065/WX FV03SM BR/TA 02 /WV 23538KT/TB LGT/ IC TRACE RIME / RM TCU W-NE	
12. /RM →	Remarks: TCU W-NE

Examples

/RM TCU W-NE (Towering Cumulus West thru Northeast)



Knowledge Check D

REVIEW what you have learned so far about PIREPs. ANSWER each question listed below.

1. Which example represents wind reported on a PIREP? *(Select the correct answer.)*
 - ☐ /SK BKN-TOP065
 - ☐ /WX FV03SM BR
 - ☒ /WV 23538KT
2. What are the abbreviations used to describe turbulence in a PIREP? *(Select all correct answers that apply.)*
 - ☒ MOD
 - ☒ EXTRM
 - ☐ EXT
 - ☒ SEV
 - ☒ LGT
 - ☐ SV
3. Which of these types of information can be found on a PIREP? *(Select all correct answers that apply.)*
 - ☒ Aircraft type
 - ☒ Location
 - ☒ Remarks
 - ☒ Altitude
 - ☒ Icing
 - ☒ Temperature

PIREP Recording and Decoding Summary

A pilot operates in an environment that is unique only to them, and weather can take a turn for the worse in an instant. When a pilot issues a pilot weather report, there is usually hazardous weather present. You must communicate the information rapidly and accurately to surrounding facilities in order help other aircraft coming into the vicinity know what to expect or what to avoid.

Report 2

PIREP FORM	
Pilot Weather Report → Space Symbol	
3-Letter SA Identifier O K C <i>"Oak City"</i> 1. UA → Routine Report UUA → Urgent Report	
2. /OV →	Location: OKC
3. /TM →	Time: 1114 Z
4. /FL →	Altitude/Flight Level: 070 - 075
5. /TP →	Aircraft Type: <i>Citation? C550</i>
Items 1 through 5 are mandatory for all PIREP's	
6. /SK →	Sky Cover:
7. /WX →	Flight Visibility and Weather:
8. /TA →	Temperature (Celsius): 04
9. /WV →	Wind:
10. /TB →	Turbulence: <i>MOD CHOP 030</i>
11. /IC →	Icing:
12. /RM →	Remarks: <i>LLWS +/- 20 KTS ON FINAL</i>
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Mod to Turb. (chop)

Report 3

PIREP FORM	
Pilot Weather Report 3-Letter SA Identifier	
Z <u>Ft Worth Center</u> F W →	1. UA → <u>UUA</u> → <u>UUA</u> → Routine Report Urgent Report
2. /OV →	Location: OKC
3. /TM →	Time: 1123Z
4. /FL →	Altitude/Flight Level: SFC - 150 Surface
5. /TP →	Aircraft Type: Cessna 40 Beech 800
Items 1 through 5 are mandatory for all PIREP's	
6. /SK →	Sky Cover: OVC 017 - TOP 090 / OVC 110 - TOP
7. /WX →	Flight Visibility and Weather:
8. /TA →	Temperature (Celsius):
9. /WV →	Wind:
10. /TB →	Turbulence:
11. /IC →	Icing:
12. /RM →	Remarks:
FAA Form 7110-2 (1-85) Supersedes Previous Edition	

SUMMARY

This module introduced you to the PIREP Program operated by the National Weather Service (NWS) and the Federal Aviation Administration (FAA). You learned to solicit, record, handle, and decode PIREPs.

In accordance with FAA Orders JO 7110.65, Air Traffic Control, and JO 7110.10, Flight Services; the Aeronautical Information Manual (AIM); FAA-H-8083-28, Aviation Weather Handbook; and FAA Form 7110-2 (PIREP), you should now be able to:

- Identify purpose and uses of a PIREP
- Identify content included in a PIREP
- Identify information recorded in a PIREP

KNOWLEDGE CHECK ANSWER KEY

Knowledge Check A

1. Severe icing, Volcanic ash, Low-level wind shear, Hail; 2. ATCTs and TRACONS

Knowledge Check B

1. Ceiling at or below 5,000, Visibility of 5 miles or less; 2. Relay pertinent PIREP information to concerned aircraft in a timely manner; 3. Report type, Location, Time, 3- or 4-letter station identifier, Type aircraft

Knowledge Check C

1. UA; 2. /TM 1600; 3. /SK BKN-TOP065

Knowledge Check D

1. /WV 23538KT; 2. MOD, EXTRM, SEV, LGT; 3. Aircraft type, Location, Remarks, Altitude, Icing, Temperature